

MACHINE LEARNING FOR STROKE PREDICTION: CAN WE DETECT STROKE BEFORE IT STRIKES ?

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- References
- Soriano, F. (2021). *Stroke Prediction Dataset*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>
 - World Health Organization (WHO). (2024). *Global Health Estimates: Stroke Mortality and Disability Statistics*. Retrieved from <https://www.who.int>
 - Centers for Disease Control and Prevention (CDC). (2024). *Stroke Facts and Statistics*. Retrieved from <https://www.cdc.gov/stroke>
 - American Heart Association (AHA). (2024). *Heart Disease and Stroke Statistics—2024 Update*. Retrieved from <https://www.heart.org>

A. INTRODUCTION

WHY STROKE PREDICTION MATTERS

- Global Impact:** Millions of deaths annually (WHO, 2024)
- Economic Burden:** \$56 billion in healthcare costs (CDC, 2024)
- 87% of strokes are **ischemic** → Early prediction = better intervention (AHA, 2024)
- Traditional risk models rely on broad guidelines → Need personalized ML-based risk assessment

DATASET & CHALLENGES

Dataset: 4,909 patients

Features: gender, age, hypertension, heart disease, glucose levels, BMI, smoking status, marital status, work type, and residence type

Challenge 1: Privacy Constraints:

- HIPAA, GDPR → Most stroke datasets are institution-specific
- Limited large-scale data sharing, hospitals rely on localized patient data

Challenge 2: Class Imbalance: Stroke cases account for a small % of total population → Models must handle imbalance while staying interpretable

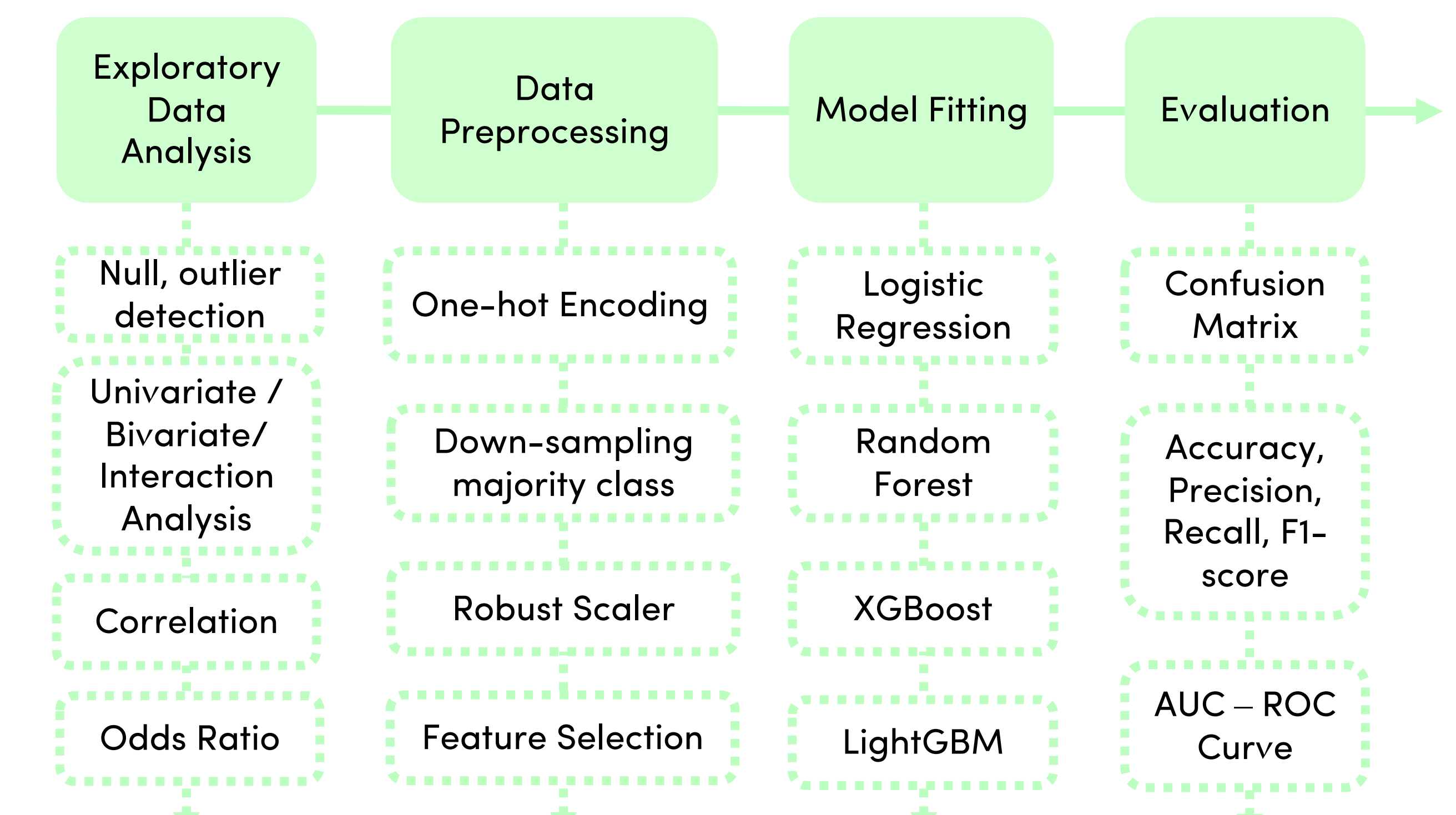
MODEL SELECTION & JUSTIFICATION

- Logistic Regression** → Baseline, easy to interpret
- Random Forest** → Captures non-linear relationships, feature importance
- LightGBM & XGBoost** → Handles class imbalance, high accuracy, scalable

GOALS

- Build ML models that perform well on **small, privacy-constrained datasets** without overfitting
- Ensure **interpretability** so healthcare providers can understand and trust predictions
- Handle **class imbalance** to improve stroke detection while minimizing false negatives
- Optimize **computational efficiency** for practical hospital deployment.
- Use Logistic Regression as a baseline to **assess if complex models justify their cost and accuracy trade-offs**

B. METHODS



- Data Imbalance:** Stroke patients make up **4%** of the dataset, requiring **down-sampling to address class imbalance**.
- Adjusting Classification Threshold:** Classification thresholds adjusted to **improve detection and minimize false negatives**.
- Feature Selection:** 2 choices - full dataset and reduced dataset.

C. INSIGHTS

STROKE DOESN'T JUST TARGET THE OLD—IT TARGETS THE UNPREPARED. 20% at risk by 80+, self-employed hit hardest, and former smokers? More vulnerable than current ones. Heart disease 5.24x, hypertension 4.44x. High glucose warns you. BMI won't protect you. EARLY DETECTION IS POSSIBLE—OUR MODEL SHOWS HOW.

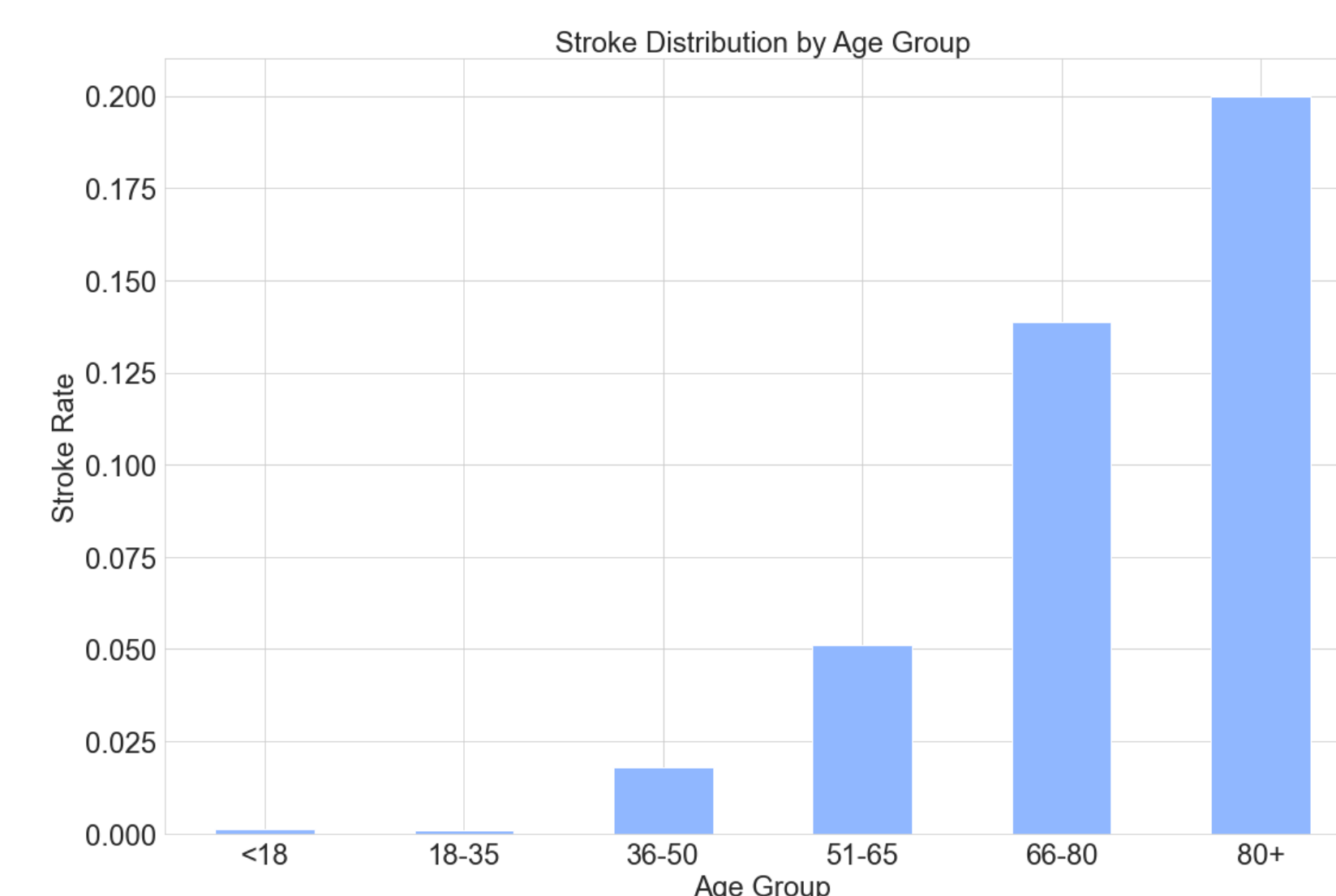


Figure 1. Stroke Distribution by Age Group

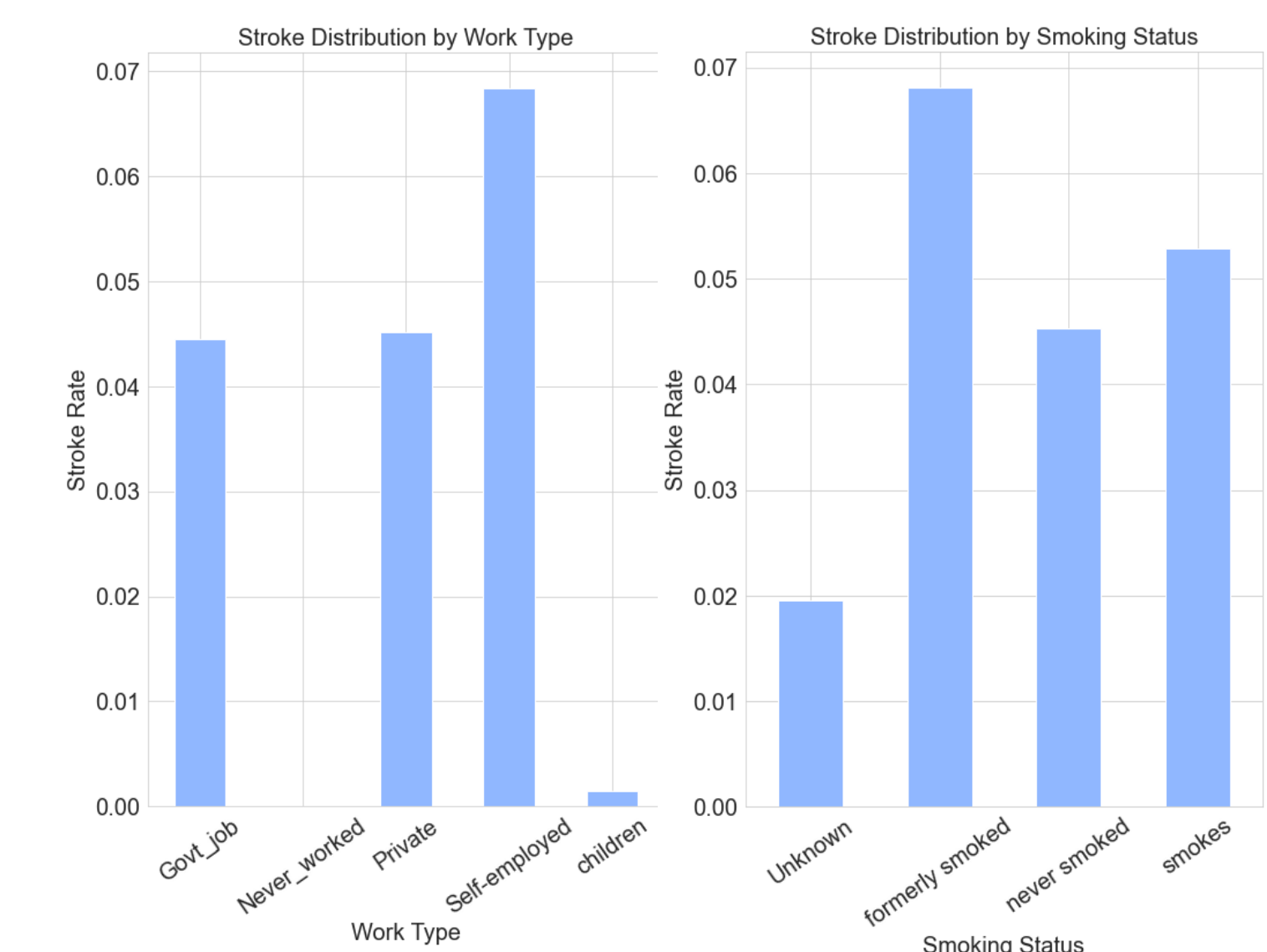


Figure 2. Stroke Distribution by Work Type

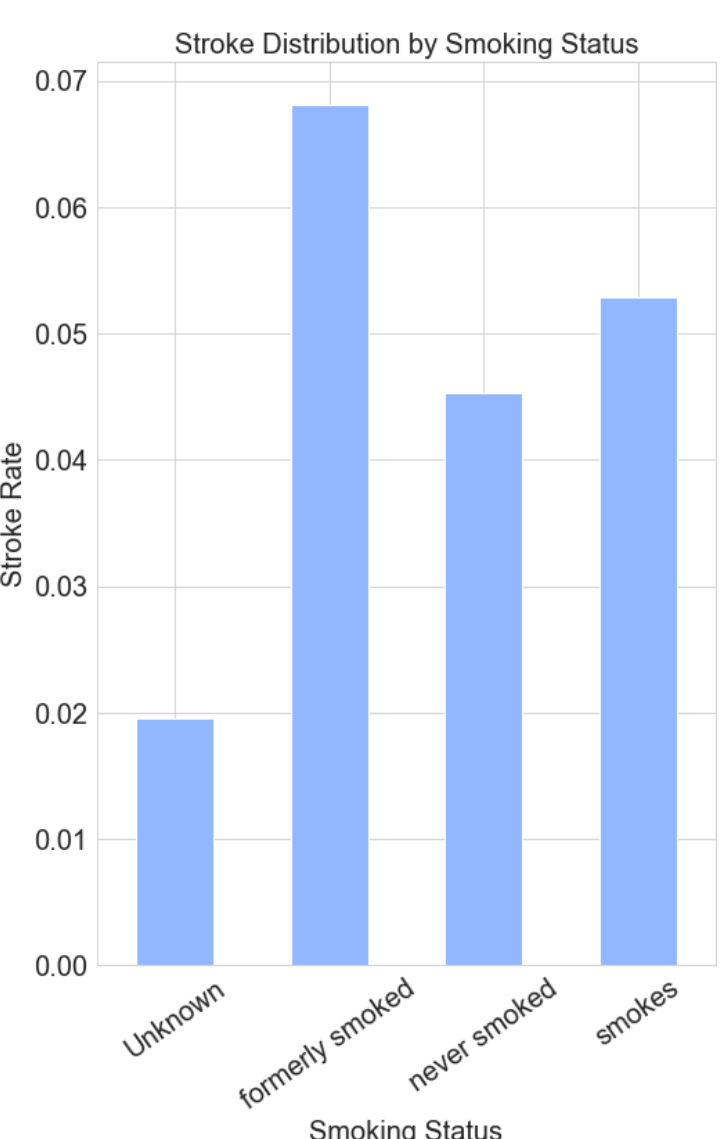


Figure 3. Stroke Distribution by Smoking Status

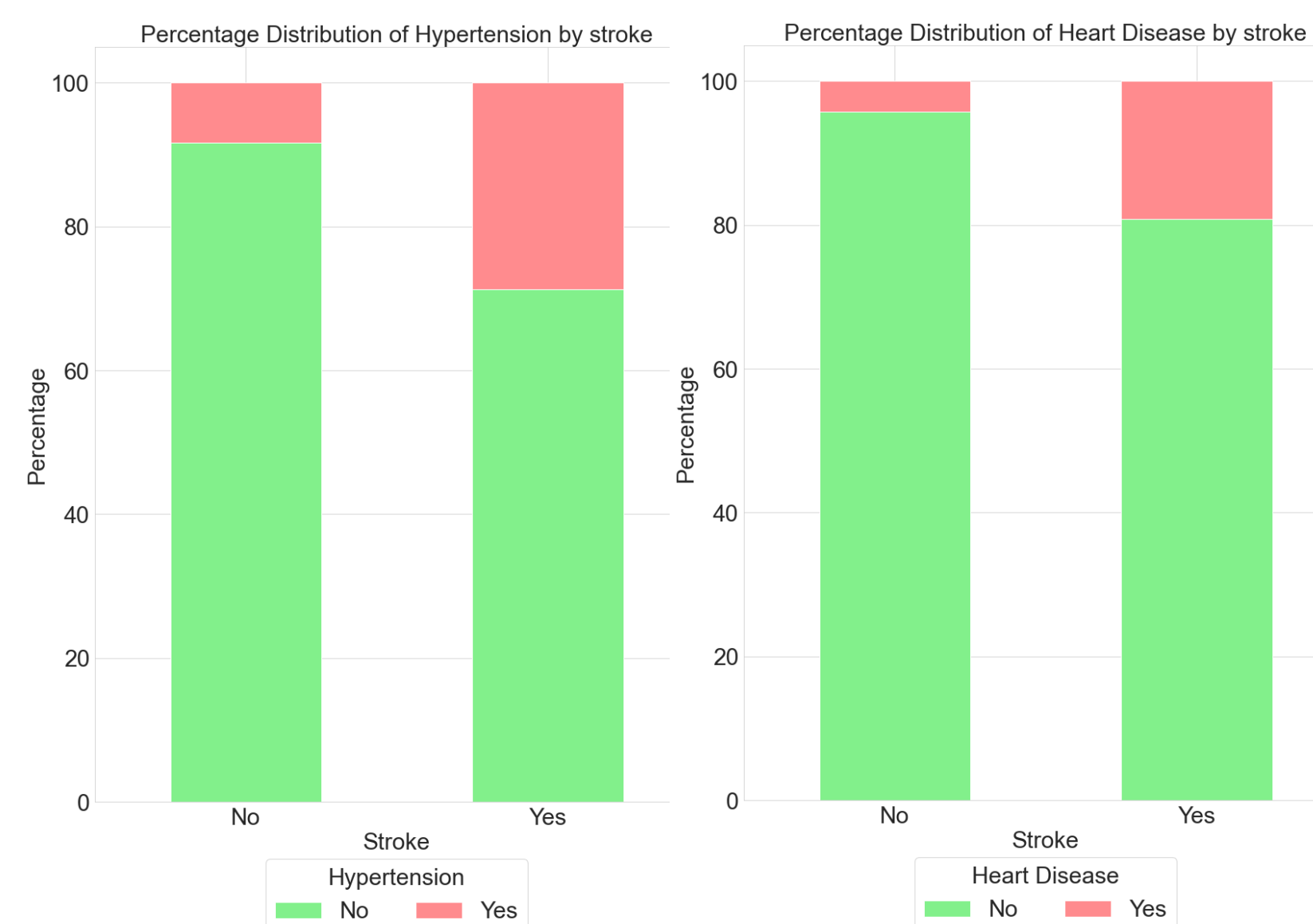


Figure 4. Percentage Distribution of Hypertension by Stroke

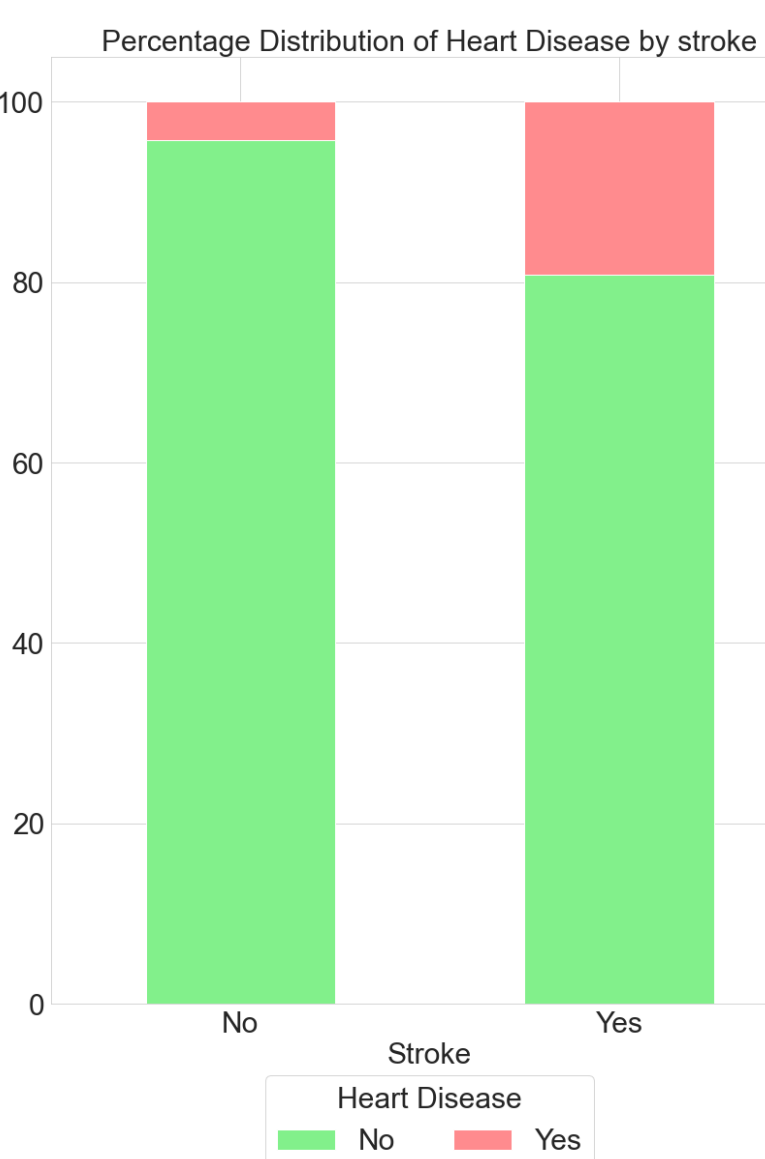


Figure 5. Percentage Distribution of Heart Disease by Stroke

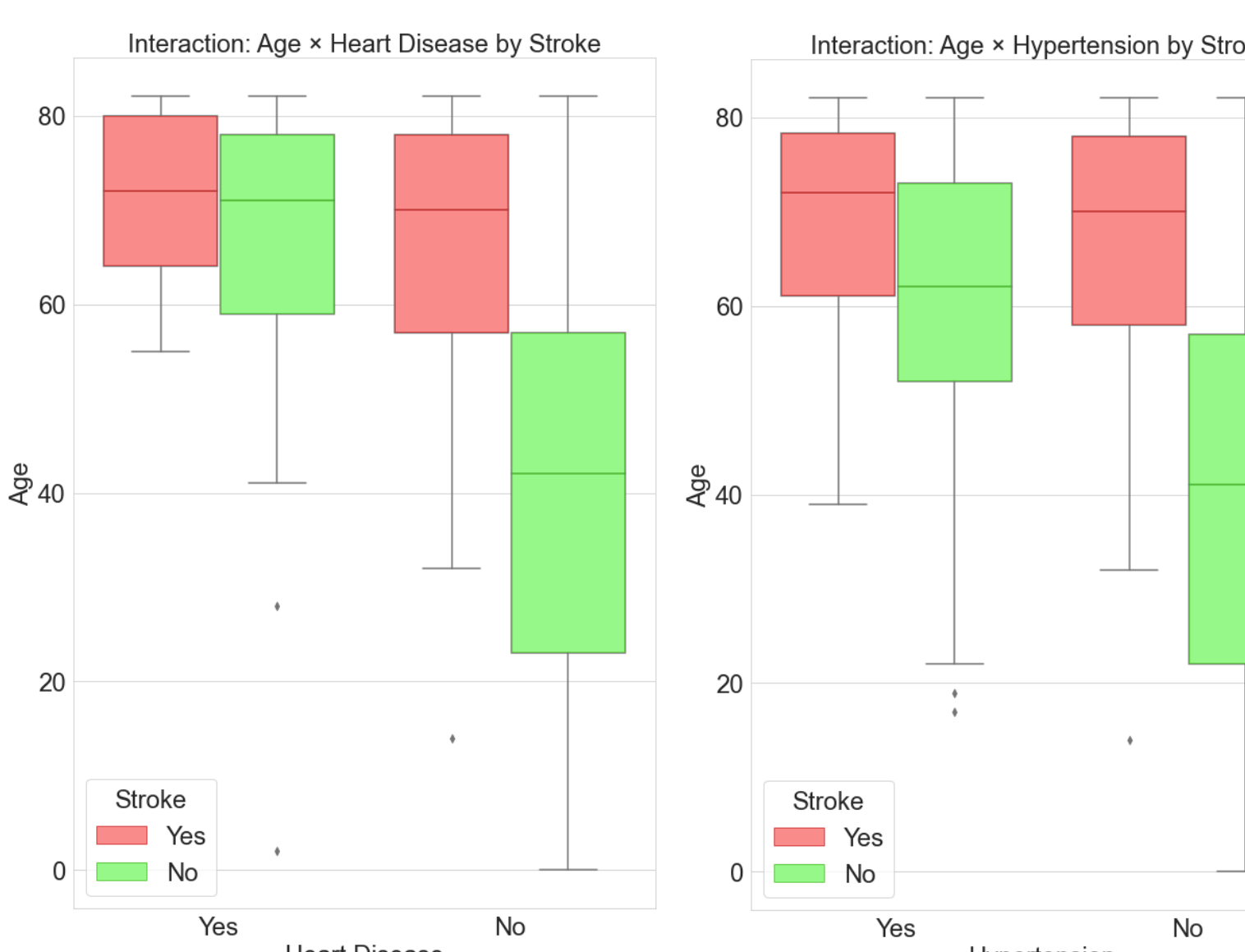


Figure 6. Interaction: Age x Heart Disease by Stroke

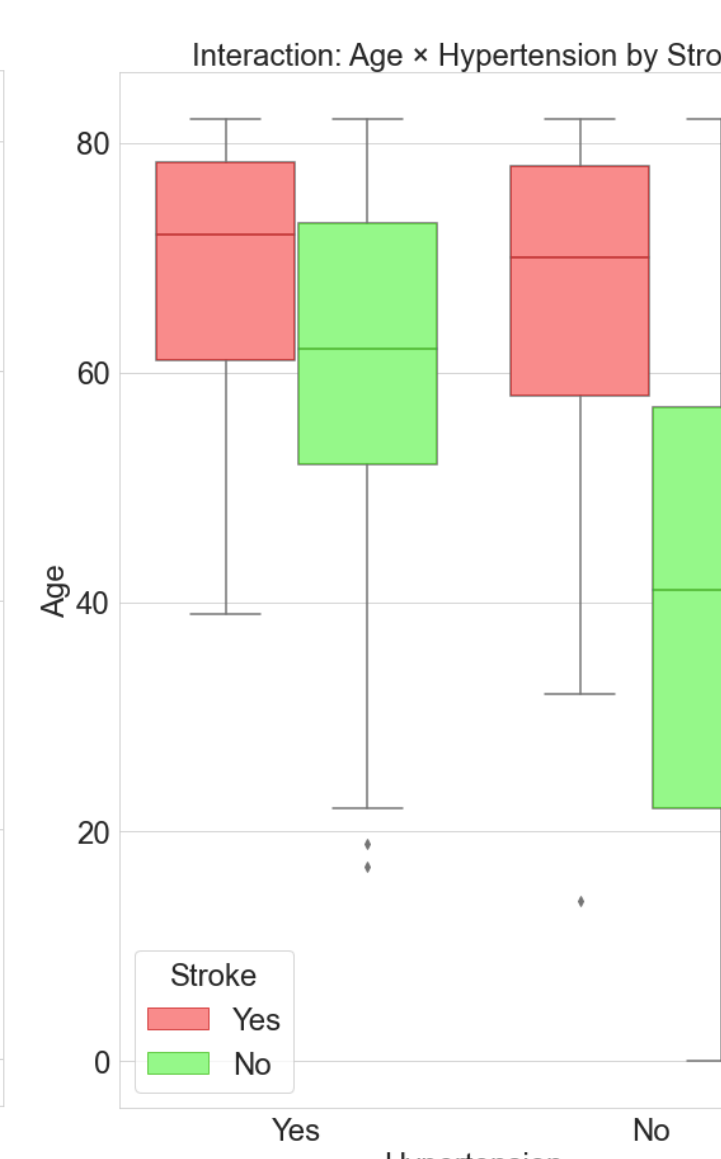


Figure 7. Interaction: Age x Hypertension by Stroke

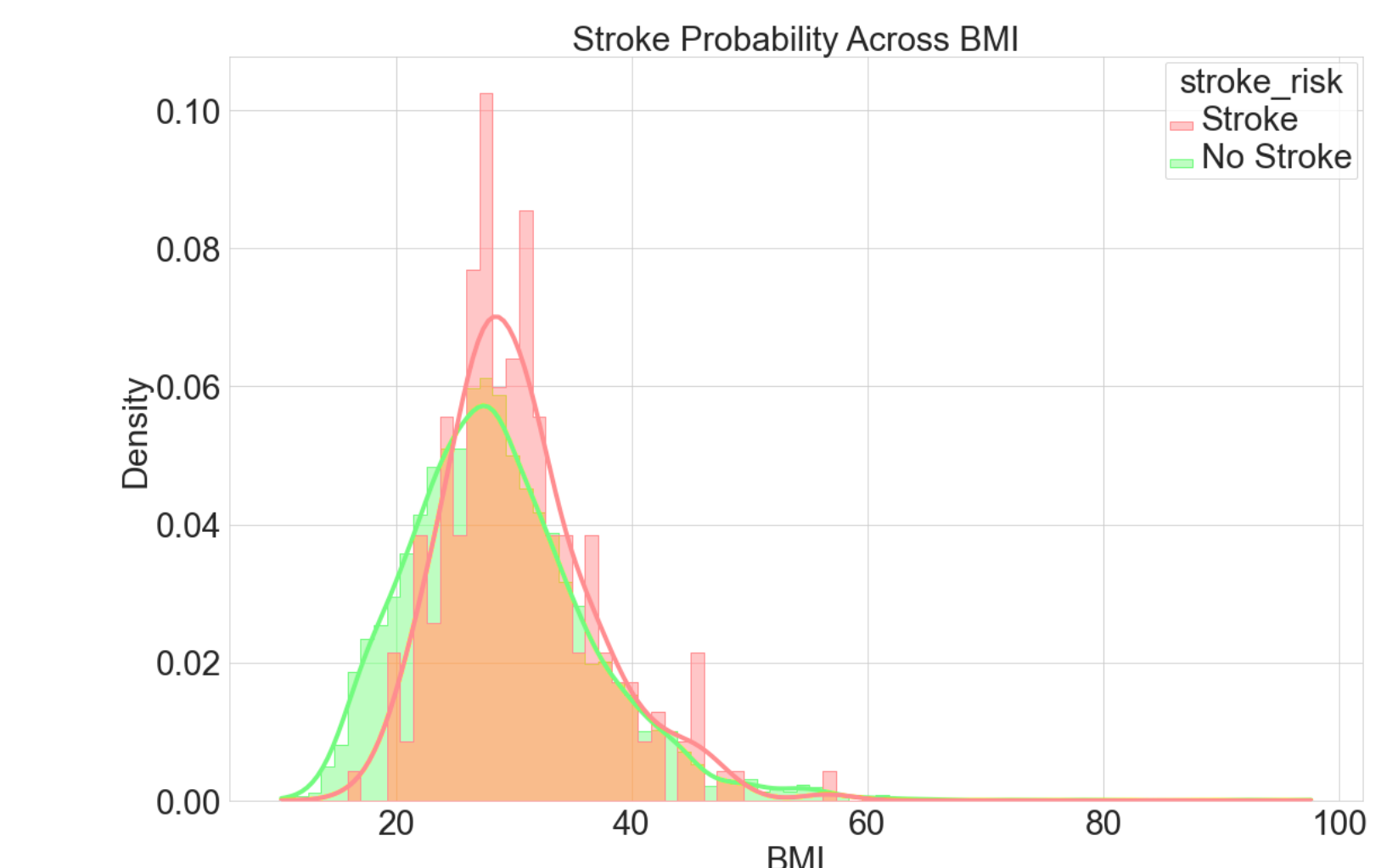


Figure 8. Stroke Probability across BMI

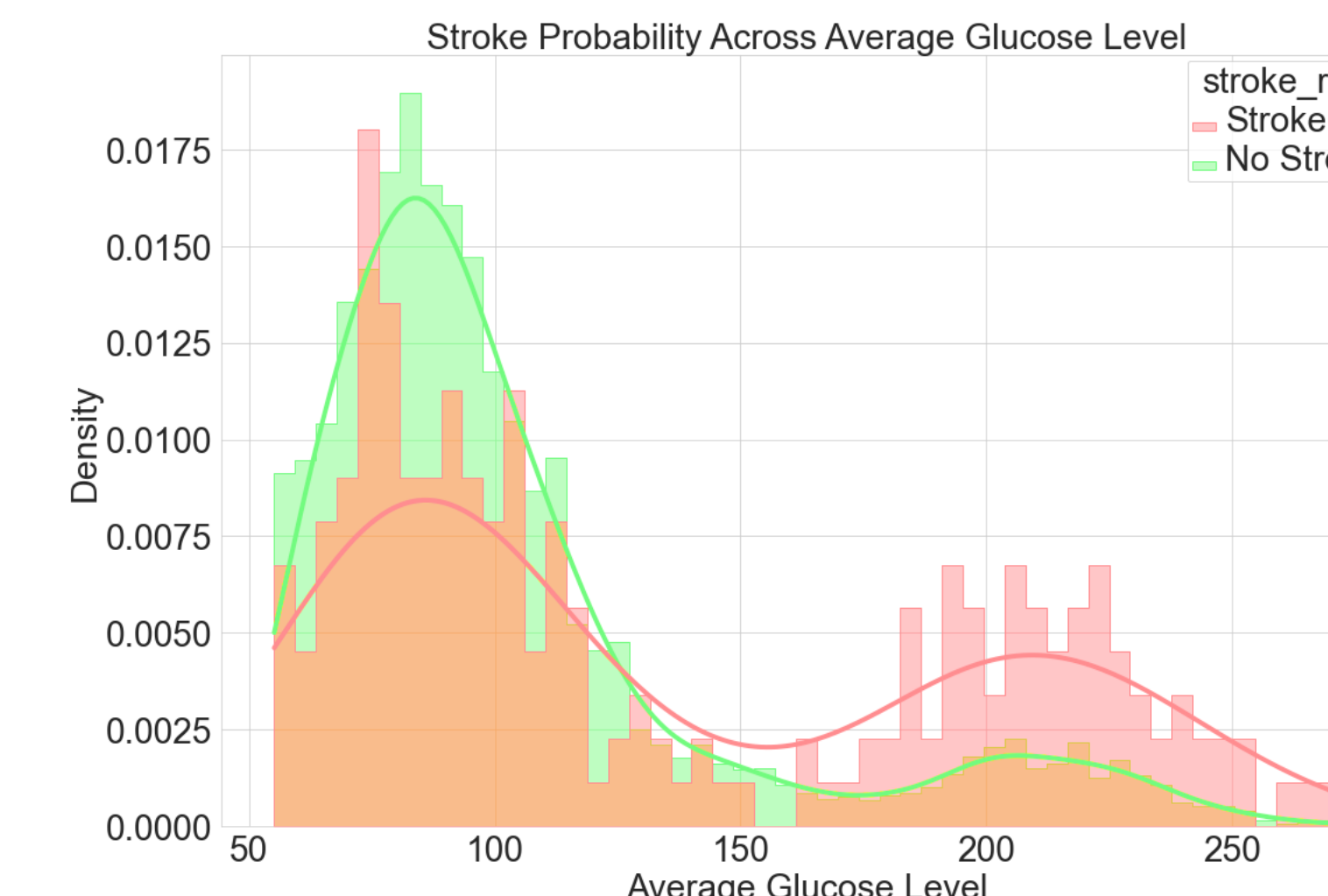


Figure 9. Stroke Probability across Average Glucose Level

D. FINAL MODEL & DISCUSSION

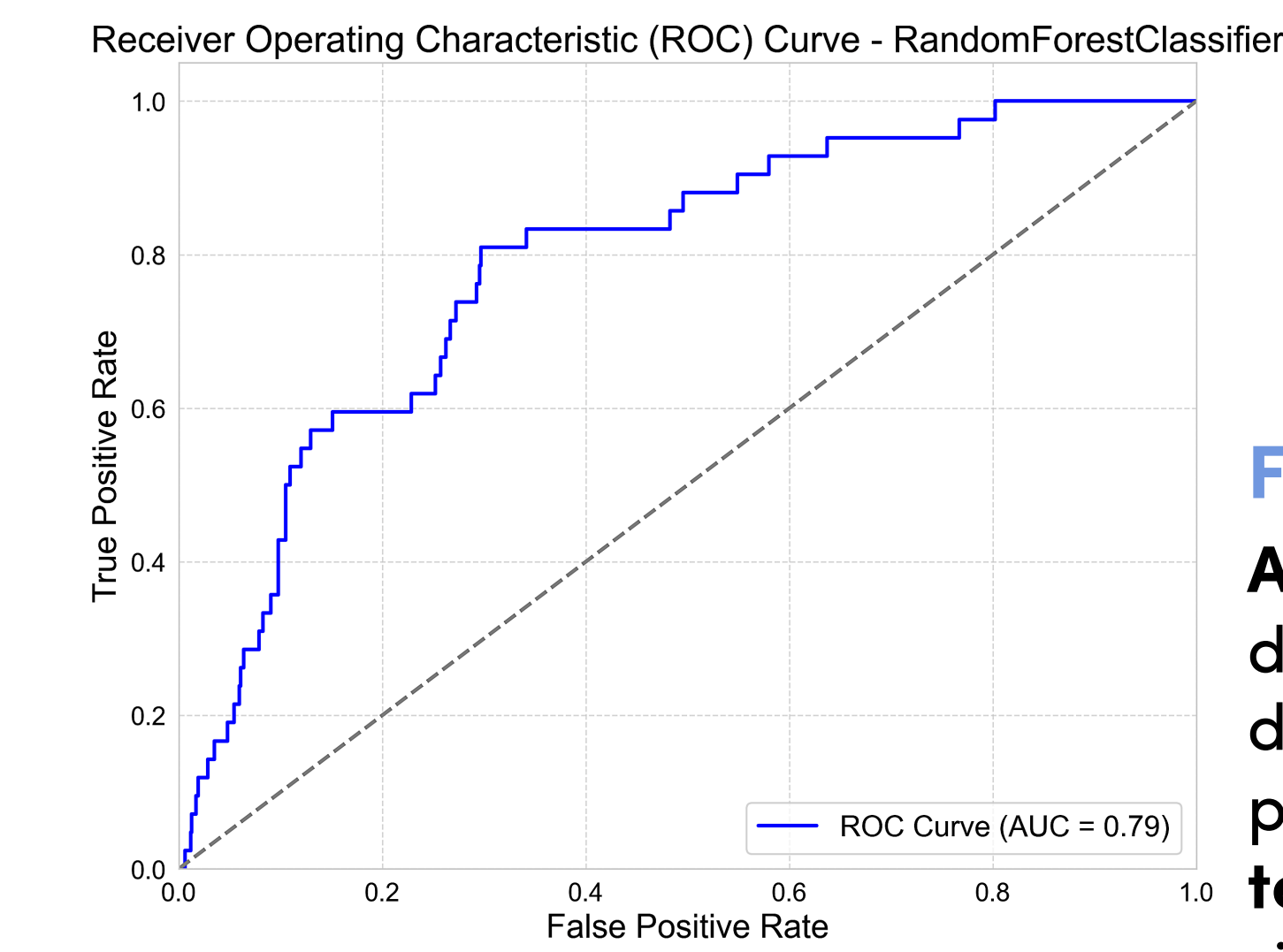


Figure 10. AUC - ROC Curve - Random Forest

MODEL	Random Forest
DATASET	Full
THRESHOLD	0.4

Table 1. Chosen model

FEATURE IMPORTANCE

Age is the most influential predictor. Metabolic factors, including **glucose level** and **BMI**, also play key roles, along with **hypertension**, a known cardiovascular risk.

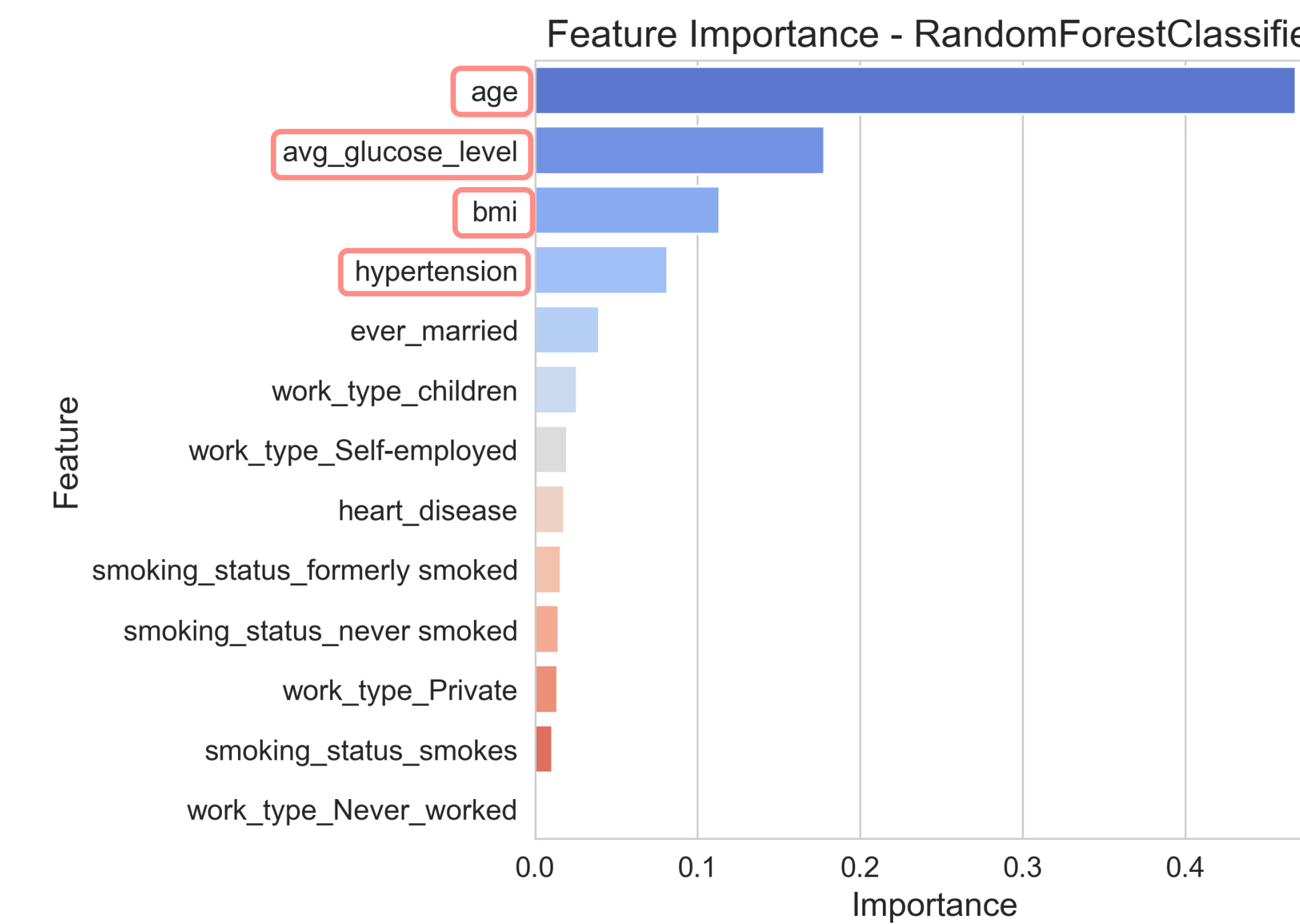


Figure 12. Feature Importance - Random Forest

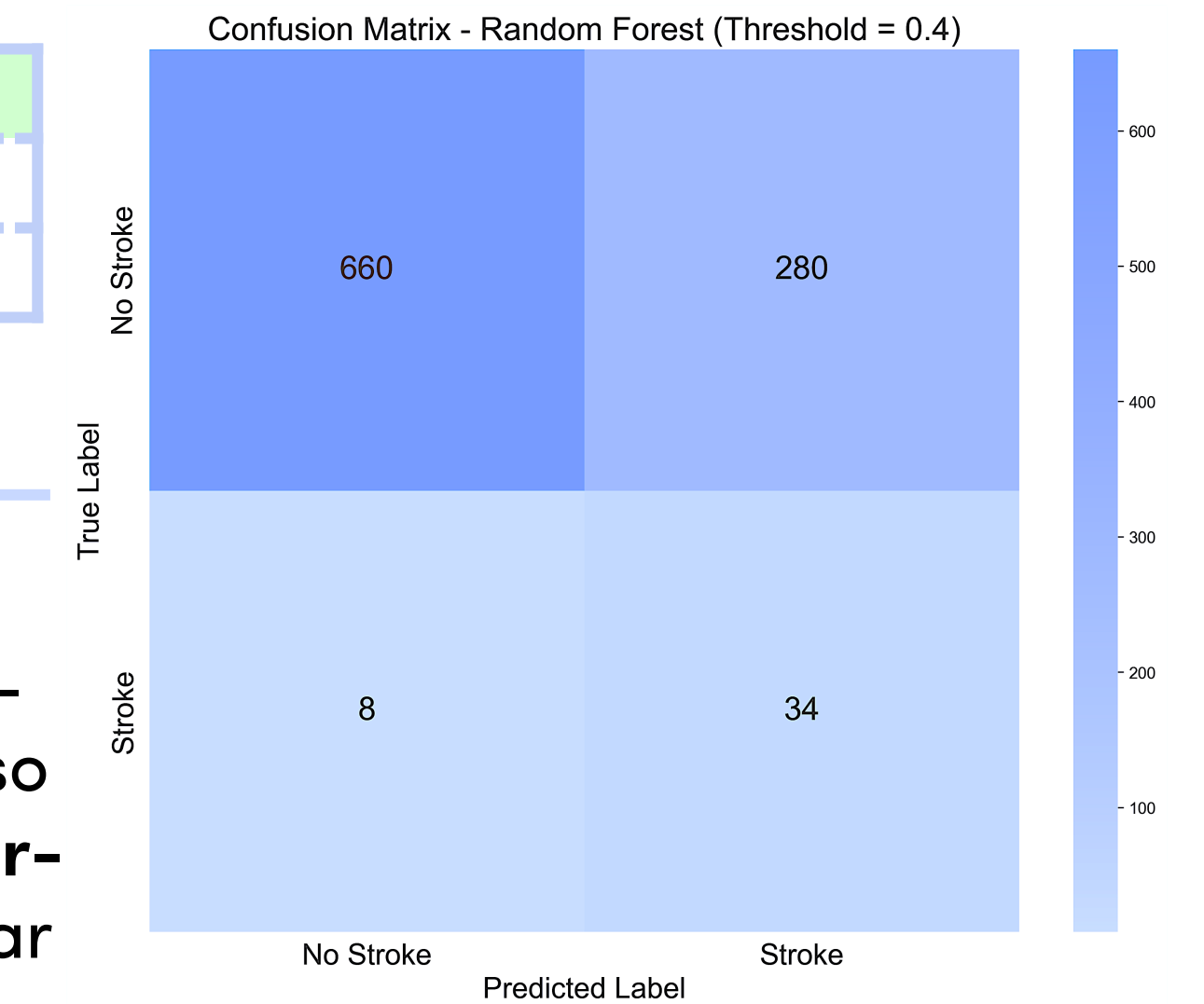


Figure 11. Confusion Matrix

DISCUSSION

Minimizing false negatives is our top priority due to the severe health risks of missed stroke cases. Since **false negatives are more dangerous than false positives**, our model **prioritizes recall** to ensure stroke cases are detected, even at the cost of some false alarms.

The model was **chosen based on a careful balance between precision and recall**. Random Forest, LightGBM, and XGBoost **perform comparable and significantly better** than Logistic Regression, **justifying the accuracy-cost trade-off**.

In practice, challenges can be **mitigated by using more balanced datasets and improving stroke case representation**.

E. DEPLOYMENT & LIFE CYCLE MANAGEMENT

- Regularly evaluates the model with **new patient data** to maintain accuracy.
- Incorporates **new risk factors** for better prediction.

SCALABILITY & FUTURE EXPANSION

MODEL DEPLOYMENT & INTEGRATION

PERFORMANCE MONITORING & CONTINUOUS IMPROVEMENT

REGULATORY COMPLIANCE & INTERPRETABILITY

- Designed to **adapt to increasing hospital data** for long-term use.
- Explores **Federated Learning** to enhance performance across hospitals **without sharing raw patient data**.

- Implemented in **hospital EHR systems** for stroke risk assessment.
- Can be extended to **wearable health devices** for proactive monitoring.

- Ensures **HIPAA, GDPR** compliance for secure patient data handling.
- Maintains **model transparency** for clinical decision-making.